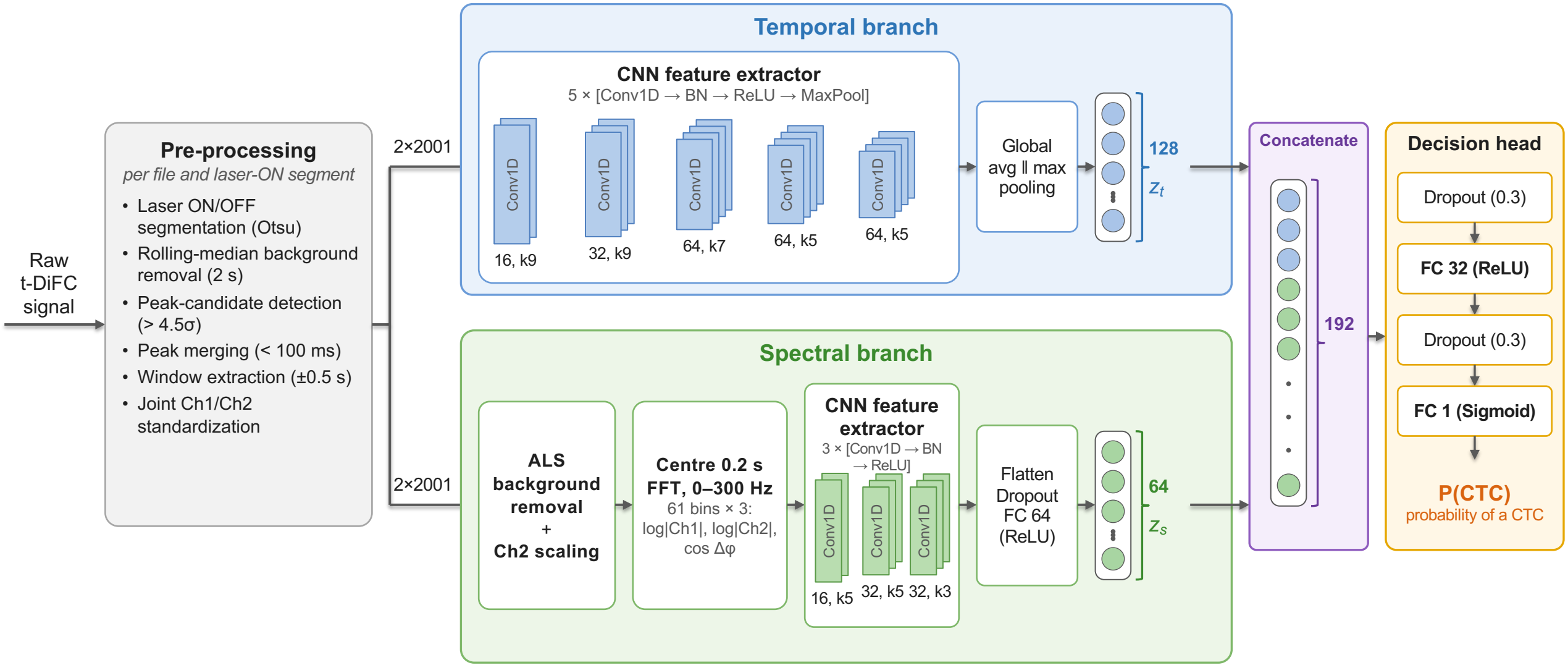

Raw t-DiFC signal
Pre-processing
per file and laser-ON segment
• Laser ON/OFF segmentation (Otsu)
• Rolling-median background removal (2 s)
• Peak-candidate detection (> 4.5σ)
• Peak merging (< 100 ms)
• Window extraction (±0.5 s)
• Joint Ch1/Ch2 standardization
2×2001
2×2001
Temporal branch
CNN feature extractor
5 × [Conv1D → BN → ReLU → MaxPool]
Conv1D
16, k9
Conv1D
32, k9
Conv1D
64, k7
Conv1D
64, k5
Conv1D
64, k5
Global avg ‖ max pooling
128
z_t
Spectral branch
ALS background removal + Ch2 scaling
Centre 0.2 s FFT, 0–300 Hz
61 bins × 3: log|Ch1|, log|Ch2|, cos Δφ
CNN feature extractor
3 × [Conv1D → BN → ReLU]
Conv1D
16, k5
Conv1D
32, k5
Conv1D
32, k3
Flatten
Dropout
FC 64 (ReLU)
64
z_s
Concatenate
192
Decision head
Dropout (0.3)
FC 32 (ReLU)
Dropout (0.3)
FC 1 (Sigmoid)
P(CTC)
probability of a CTC